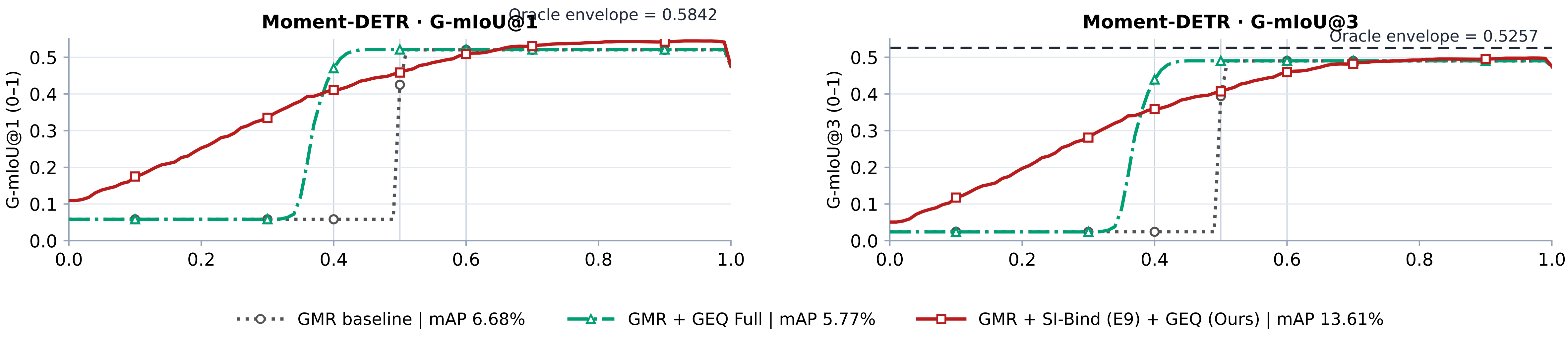
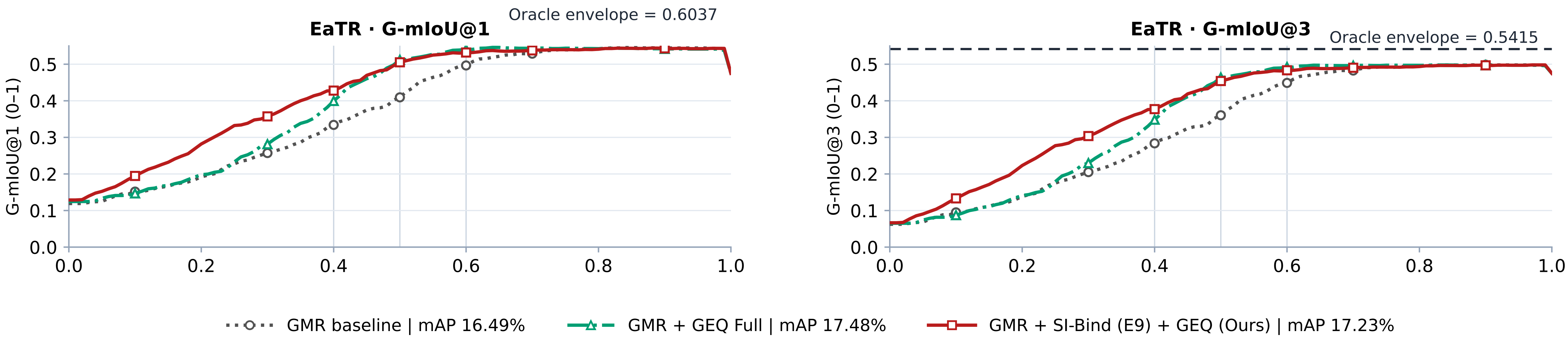
SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E9 / CAB) + GEQ

Standard-trained, fixed Val-selected best checkpoints → Standard Test (1,036 queries)

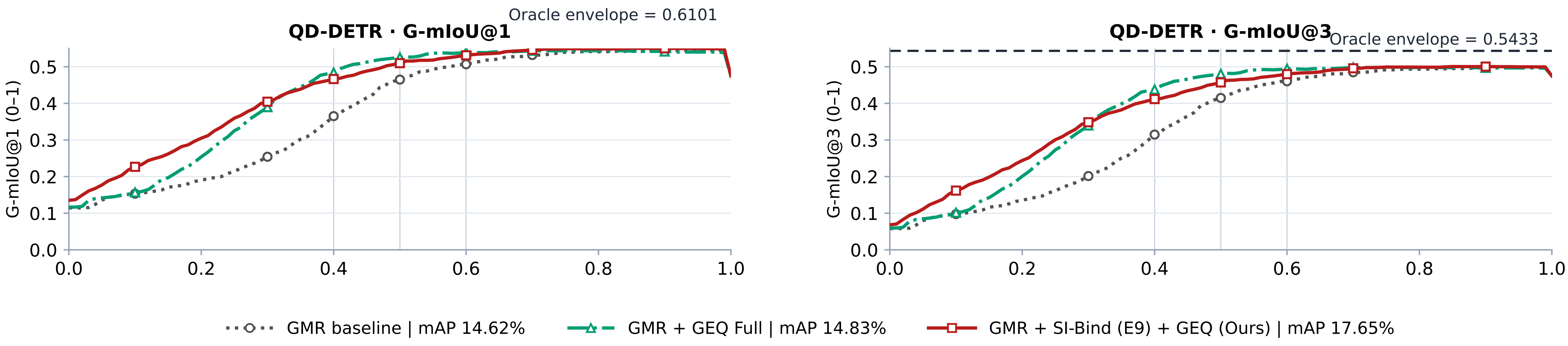
Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]



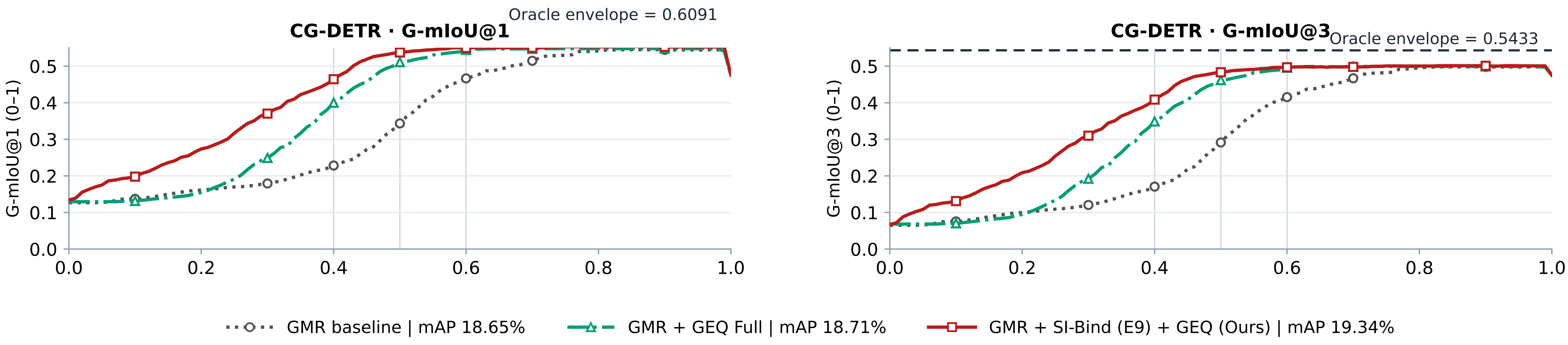
ΔmAP vs GMR: +6.93 pp (vs GEQ: +7.84 pp) **ΔG1 mean vs GMR (.4/.5/.6): +12.48 pp** **ΔG3 mean vs GMR (.4/.5/.6): +10.57 pp**



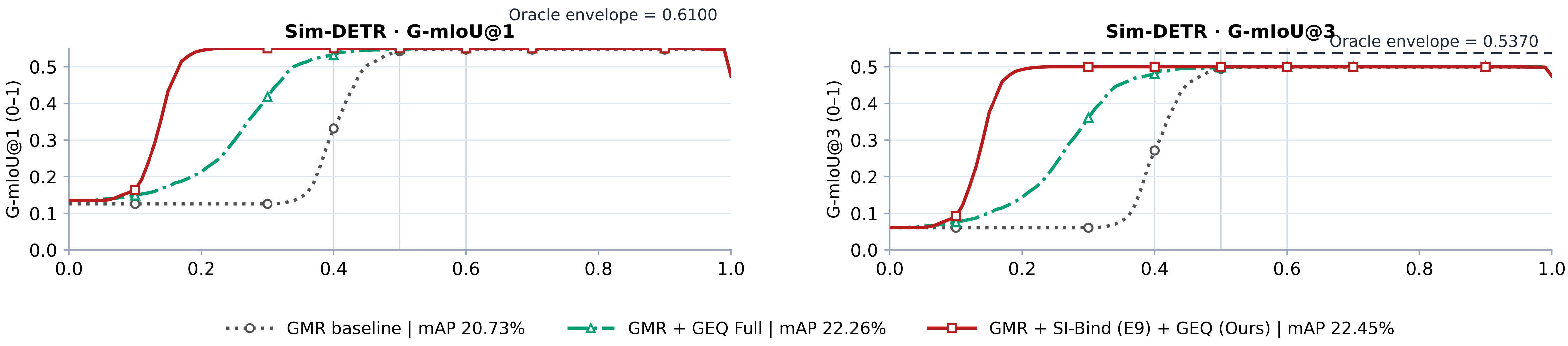
ΔmAP vs GMR: +0.74 pp (vs GEQ: -0.25 pp) **ΔG1 mean vs GMR (.4/.5/.6): +7.50 pp** **ΔG3 mean vs GMR (.4/.5/.6): +7.39 pp**



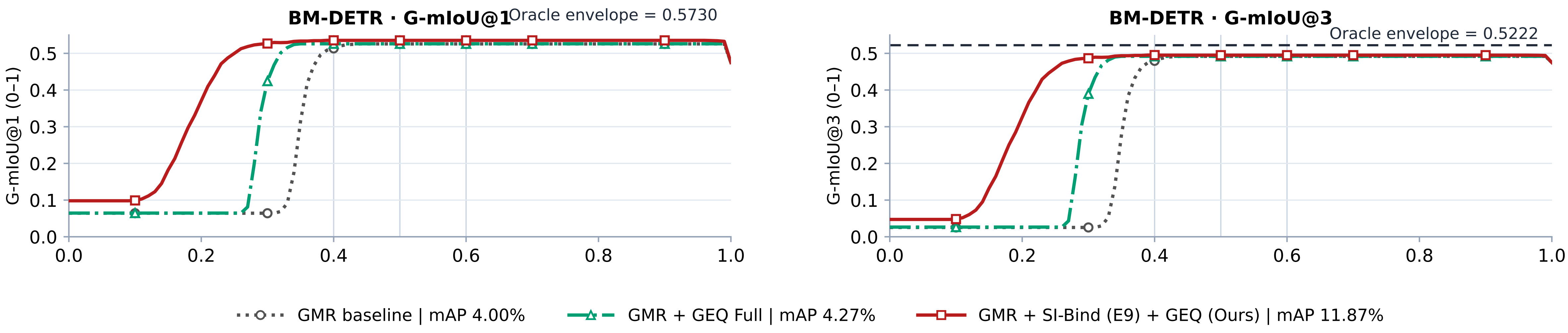
ΔmAP vs GMR: +3.03 pp (vs GEQ: +2.82 pp) **ΔG1 mean vs GMR (.4/.5/.6): +5.69 pp** **ΔG3 mean vs GMR (.4/.5/.6): +5.30 pp**



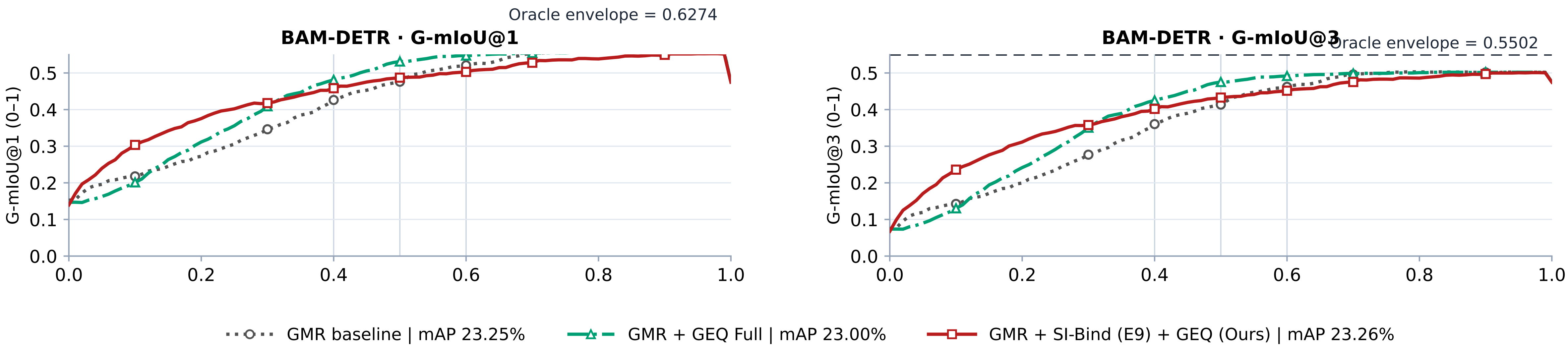
ΔmAP vs GMR: +0.69 pp (vs GEQ: +0.63 pp) **ΔG1 mean vs GMR (.4/.5/.6): +17.10 pp** **ΔG3 mean vs GMR (.4/.5/.6): +17.04 pp**



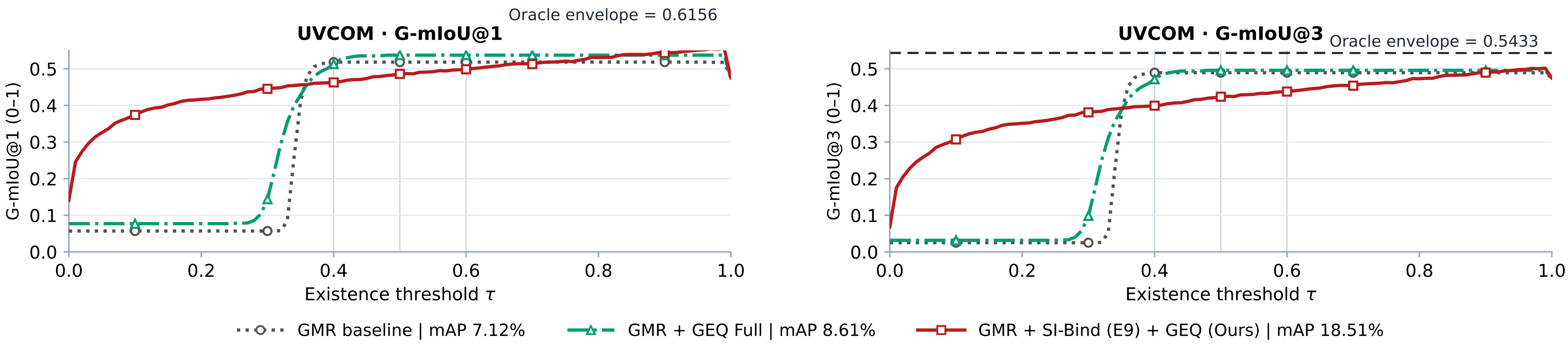
ΔmAP vs GMR: +1.72 pp (vs GEQ: +0.19 pp) **ΔG1 mean vs GMR (.4/.5/.6): +7.65 pp** **ΔG3 mean vs GMR (.4/.5/.6): +7.83 pp**



ΔmAP vs GMR: +7.87 pp (vs GEQ: +7.60 pp) **ΔG1 mean vs GMR (.4/.5/.6): +1.38 pp** **ΔG3 mean vs GMR (.4/.5/.6): +0.74 pp**



ΔmAP vs GMR: +0.01 pp (vs GEQ: +0.26 pp) **ΔG1 mean vs GMR (.4/.5/.6): +0.84 pp** **ΔG3 mean vs GMR (.4/.5/.6): +1.67 pp**



ΔmAP vs GMR: +11.39 pp (vs GEQ: +9.90 pp) **ΔG1 mean vs GMR (.4/.5/.6): -3.58 pp** **ΔG3 mean vs GMR (.4/.5/.6): -6.91 pp**

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E9) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.